



$$\frac{\cancel{x^3}}{\cancel{x^3}} \left(\frac{-4 + \frac{3}{\cancel{x}} + \frac{1}{\cancel{x^3}}}{2 + \frac{1}{\cancel{x^2}} + \frac{2}{\cancel{x^3}}} \right) = (-2)$$

$$\lim_{x \rightarrow 0} \frac{\sin(10x)}{x^3 + 10x} = \lim_{x \rightarrow 0} \left(\frac{\sin 10x}{10x} \right) \frac{10x}{x^3 + 10x} =$$

$$\lim_{x \rightarrow 0} \frac{\cancel{\sin 10x}}{\cancel{10x}} \lim_{x \rightarrow 0} \frac{10x}{x^3 + 10x} =$$

$$1 \quad \lim_{x \rightarrow 0} \frac{10}{x^2 + 10} = 1$$

$$\lim_{x \rightarrow 1} \frac{x^3 + x^2 - 5x + 3}{x^3 - 4x^2 + 5x - 2} = \lim_{x \rightarrow 1} \frac{(3x^2 + 2x - 5)}{3x^2 - 4x + 5}$$

$$\frac{-2 \pm \sqrt{4 + 60}}{6}$$

$$\begin{aligned} \frac{-2 + 8}{6} &= 1 \\ \frac{-2 - 8}{6} &= -\frac{10}{6} = -\frac{5}{3} \end{aligned}$$

$$3(x-1)\left(x + \frac{5}{3}\right) = (x-1)(3x+5)$$

$$\frac{8 \pm \sqrt{64 - 60}}{6} = \frac{\cancel{8} \mp 2}{6} = \frac{10}{6} = \frac{5}{3}$$

$$\frac{8-2}{6} = 1$$

$$3(x-1)\left(x-\frac{5}{3}\right) = (x+1)(3x-5)$$

$$\lim_{x \rightarrow 1} \frac{\cancel{(x-1)}(3x+5)}{\cancel{(x-1)}(3x-5)} = \lim_{x \rightarrow 1} \frac{3x+5}{3x-5} =$$

$$\frac{8}{-2} = \textcircled{-4}$$

$$\lim_{x \rightarrow \infty} \frac{-2x^3 + 5x^2 + 9x - 1}{3x^3 + 4x^2 + 7} =$$

$$\lim_{x \rightarrow \infty} \frac{-6x^2 + 10x + 9}{9x^2 + 8x} \rightarrow$$

$$\lim_{x \rightarrow \infty} \frac{-12x + 10}{18x + 8}$$

$$\lim_{x \rightarrow \infty} \frac{-12}{18} = -\frac{2}{3}$$

$$\lim_{x \rightarrow 0} \frac{\sin(2025x)}{\sin(2026x)} = \lim_{x \rightarrow 0} \frac{\cos(2025x) \cdot 2025}{\cos(2026x) \cdot 2026}$$

$$\lim_{x \rightarrow \infty} \frac{e^x + 2}{4x^3} = \lim_{x \rightarrow \infty} \frac{e^x}{12x^2} = \lim_{x \rightarrow \infty} \frac{e^x}{24x}$$

$$= \lim_{x \rightarrow \infty} \frac{e^x}{24} = +\infty$$

$$y = x^2$$

$$y = x^3$$

$$\sqrt{x}$$

$$\frac{1}{x}$$

$$\frac{e^x}{x^{1000000}}$$

$$\lim_{x \rightarrow \infty} \frac{e^x}{x^5}$$

$$\Rightarrow \lim_{x \rightarrow \infty} \frac{e^x}{x^n} =$$

$$\lim_{x \rightarrow \infty} \frac{e^x}{x^6}$$

$$= \lim_{x \rightarrow \infty} \frac{e^x}{n!} = +\infty$$

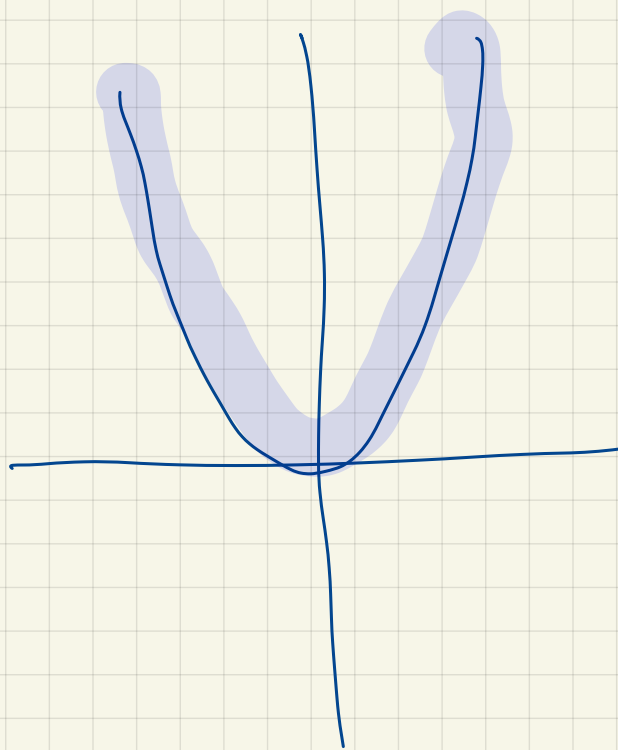
$$5! = 1 \cdot 2 \cdot 3 \cdot 4 \cdot 5$$

$$6! = 1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6$$

$$y = x^2$$

$$y'' = 2 > 0 \Rightarrow$$

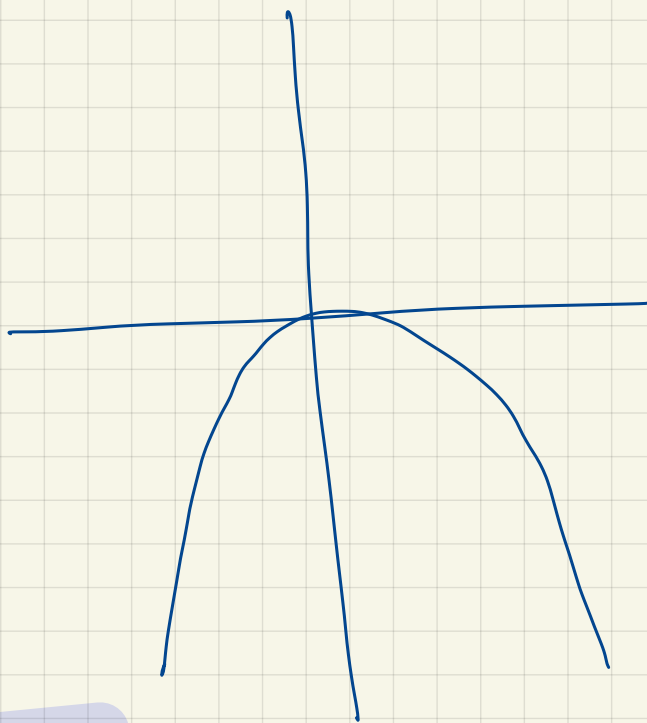
выпукла



$$y = -x^2$$

$$y'' = -2 < 0$$

выпукла



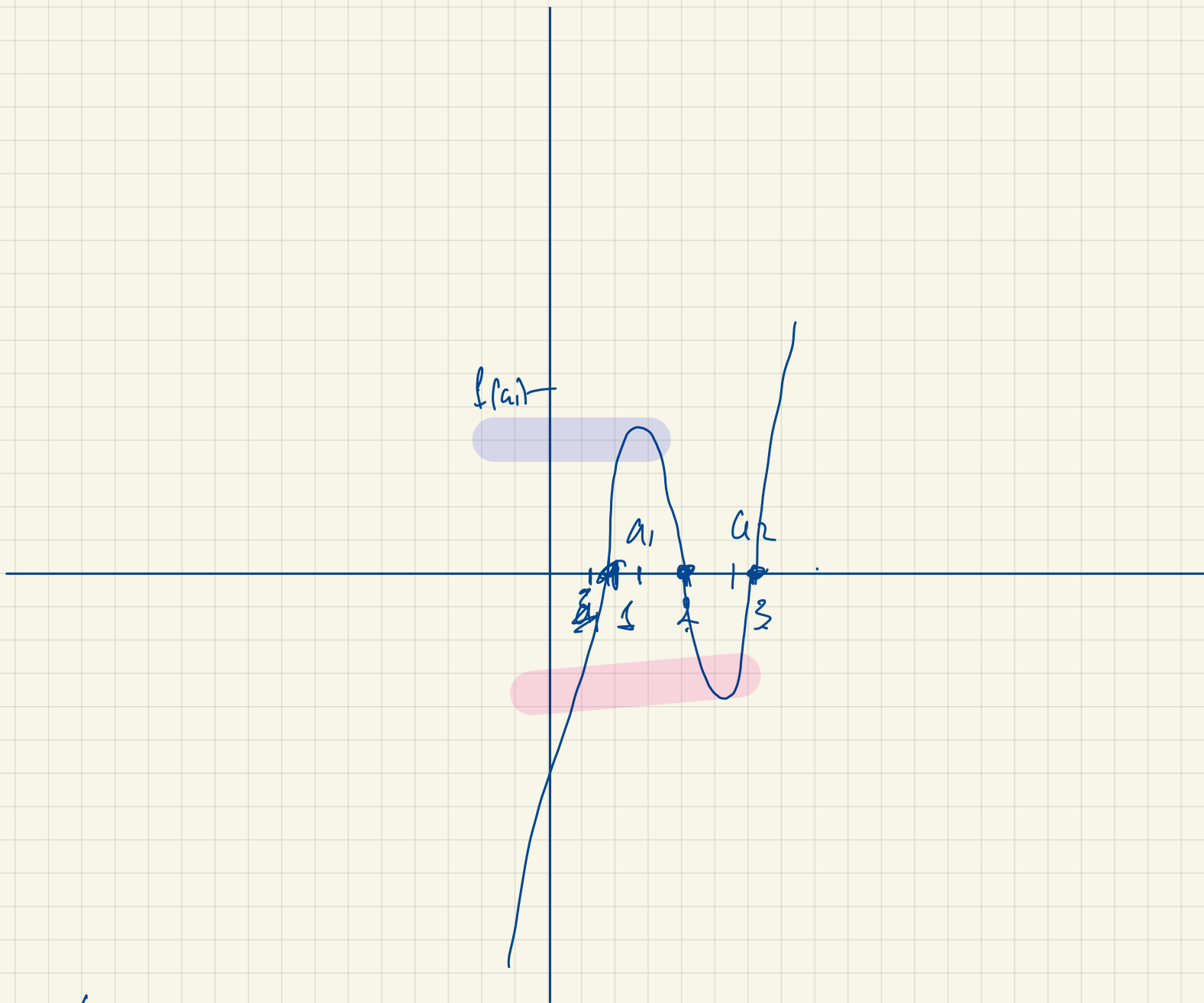
$$y'' = 0$$

условие

т. перегиба

$$y = x^3 - 6x^2 + 11x - 6$$

1, 2, 3 корни.



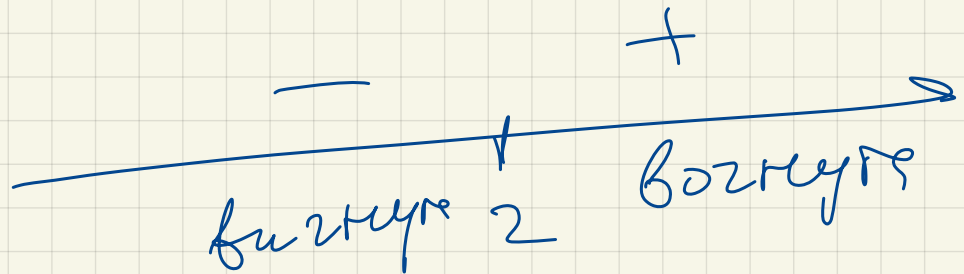
$$y'(x) = 3x^2 - 12x + 11 = 0$$

$$\frac{12 \pm \sqrt{12}}{6} = \frac{12 \pm 2\sqrt{3}}{6} = \frac{6 \pm \sqrt{3}}{3}$$

$$a_1 = \frac{6 - \sqrt{3}}{3} \quad \swarrow \quad \searrow \quad \frac{6 + \sqrt{3}}{3} = a_2$$

$$y''(x) = 6x - 12 = 0$$

$$(x = 2)$$



т. пересечения

$$y = 4x^4 - 3x^3 + 2x^2 + x + 2026$$

$$y'' = 48x^2 + 18x + 4 = 0$$

$$24x^2 + 9x + 2 = 0 \quad > 0$$

$$+81 - 4 \cdot 24 \cdot 2 < 0$$

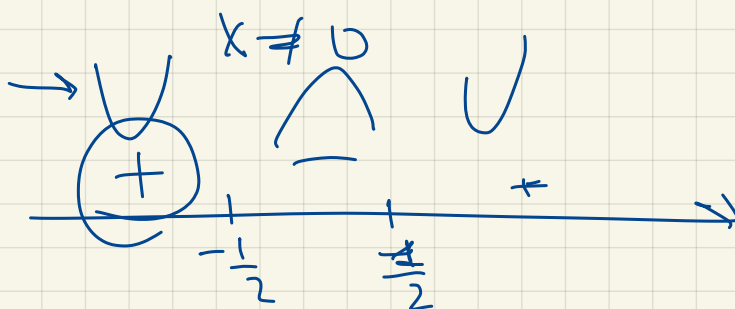
т. пересечения нет!

$$y = 2x^2 + \ln x$$

$$4x + \frac{1}{x} = 4 - \frac{1}{x^2}$$

$$\frac{4x^2 - 1}{x^2} = 0$$

$$x^2 = \frac{1}{4} \Rightarrow x = \pm \frac{1}{2}$$



$$y = \frac{x}{\ln x} \quad \text{+. representation}$$

$$y' = \frac{\ln x - \frac{1}{x} \cdot x}{\ln^2 x} = \frac{\ln x - 1}{\ln^2 x} =$$

$$y'' = \frac{\frac{1}{x} \cdot \ln^2 x - 2 \ln x \cdot \frac{1}{x} (\ln x - 1)}{\ln^4 x}$$

$$y'' = \frac{\frac{\ln^2 x}{x} - 2 \frac{\ln^2 x}{x} + \frac{\ln x}{x}}{\ln^4 x} = \frac{-\frac{\ln^2 x}{x} + \frac{\ln x}{x}}{\ln^4 x}$$

$$y = \frac{(x+2)^2 (x-1) + 2026}{x \cdot \ln^4 x} \quad (20; +\infty)?$$

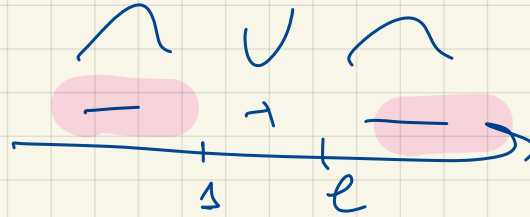
$$\ln x - \ln^2 x = 0$$

$$\ln x (1 - \ln x) = 0$$

$$\ln x = 0 \quad \ln x = 1$$

$$x = 1$$

$$x = e$$



$$y = (x+2)^2(x-1) + 2026$$

y''

$$(20; +\infty)?$$

$$y' = 2(x+2)(x-1) + (x+2)^2$$

$$y'' = 2(x-1) + 2(x+2) + 2(x+2)$$

$$2x - 2 + 2x + 4 + 2x + 4 =$$

$$6x + 6 = 0$$

$$x = -1$$

